Chemistry SW03 – Electronic configuration

1. The Four Quantum Numbers and Their Meanings

- Principal quantum number (n) →
- Azimuthal quantum number (l) →
- Magnetic quantum number (m) →
- Spin quantum number (s) →

2. What Does the Pauli Exclusion Principle State?

3. Electron Distribution of Phosphorus

Shell	n	l	Orbital Type	m	s
K					
L					
L					
L					
L					
M					
M					
M					
M					

Note: Hund's Rule applies.

4. Electron Distribution of Oxygen

Shell	n	l	Orbital Type	m	s
K					
L					
L					
L					
L					

Note: Hund's Rule applies.

5. Electron Distribution of Magnesium

Shell	n	l	Orbital Type	m	s
K					
L					
L					
L					
L					
M					

6. What Does the Heisenberg Uncertainty Principle State?

7. What Is an Atomic Orbital?

8. Which Statements Are Correct?

• When the outermost shell of an atom is completely filled with electrons, the atom is in a noble gas state. →

• The Schrödinger equation can calculate the probability of finding electrons around a nucleus. →

• Neutrons and protons always exist in a 1:1 ratio. →

• Atoms not in the noble gas state form molecules. →

9. Number of Valence Electrons (you can also do this in SW4)

• Sodium →
• Sulfur →
• Arsenic →
• Neon →

Solutions Chemistry SW03 –

1. The Four Quantum Numbers and Their Meanings

- Principal quantum number (n) → Determines the size of the orbital.
- Azimuthal quantum number (l) → Defines the shape of the orbital.
- Magnetic quantum number (m) → Describes the orientation of the orbital in space.
- Spin quantum number (s) → Represents the intrinsic spin of the electron.

2. What Does the Pauli Exclusion Principle State?

Fermions must always differ in at least one quantum number.

3. Electron Distribution of Phosphorus

Shell	n	l	Orbital Type	m	s
K	1	0	s	0	+0.5 / -0.5
L	2	0	s	0	+0.5 / -0.5
L	2	1	p	-1	+0.5 / -0.5
L	2	1	p	0	+0.5 / -0.5
L	2	1	p	1	+0.5 / -0.5
M	3	0	s	0	+0.5 / -0.5
M	3	1	p	-1	+0.5 or -0.5
M	3	1	p	0	+0.5 or -0.5
M	3	1	p	1	+0.5 or -0.5

Note: Hund's Rule applies.

4. Electron Distribution of Oxygen

Shell	n	l	Orbital Type	m	s
K	1	0	s	0	+0.5 / -0.5
L	2	0	s	0	+0.5 / -0.5
L	2	1	p	-1	+0.5 / -0.5
L	2	1	p	0	+0.5 or -0.5

L	2	1	p	1	+0.5 or -0.5

Note: Hund's Rule applies.

5. Electron Distribution of Magnesium

Shell	n	l	Orbital Type	m	s
K	1	0	s	0	+0.5 / -0.5
L	2	0	s	0	+0.5 / -0.5
L	2	1	p	-1	+0.5 / -0.5
L	2	1	p	0	+0.5 / -0.5
L	2	1	p	1	+0.5 / -0.5
M	3	0	s	0	+0.5 / -0.5

6. What Does the Heisenberg Uncertainty Principle State?

One can determine the location where an electron resides but not the time (and vice versa).

7. What Is an Atomic Orbital?

An atomic orbital represents the region where electrons are most likely to be found.

8. Which Statements Are Correct?

• When the outermost shell of an atom is completely filled with electrons, the atom is in a noble gas state. → Correct

• The Schrödinger equation can calculate the probability of finding electrons around a nucleus. → Correct

• Neutrons and protons always exist in a 1:1 ratio. → Incorrect

• Atoms not in the noble gas state form molecules. → Correct

9. Number of Valence Electrons (you can also do this in SW4)

• Sodium → 1
• Sulfur → 6
• Arsenic → 5
• Neon → none or 8 → Noble gas, octet fulfilled